Appelli di MAURIZIO GABBRIELLI

Iniziato	venerdì, 26 giugno 2020, 12:03
Stato	Completato
Terminato	venerdì, 26 giugno 2020, 12:16
Tempo impiegato	12 min. 20 secondi
Punteggio	7,00/10,00
Valutazione	**21,70** su un massimo di 31,00 (**70**%)

Domanda **1**
Risposta corretta
Punteggio ottenuto 2,00 su 2,00
Contrassegna domanda

Let f, g be the functions defined as $f(n) = 10^3 n \log n$ and $g(n) = \frac{n^2}{10^5 \log n}$.

Scegli una o più alternative:

- [ ] $f \in \Omega(g)$
- [ ] $f \in \Theta(g)$
- [x] $f \in O(g)$ ✔

Your answer is correct.

La risposta corretta è: $f \in O(g)$

Domanda **2**
Parzialmente corretta
Punteggio ottenuto 1,00 su 2,00
Contrassegna domanda

Nondeterministic Turing Machines:

Scegli una o più alternative:

- [ ] Can be simulated by deterministic TMs.
- [x] If working in polynomial time, can be used to characterize $\mathbf{NP}$ ✔
- [ ] Always work in polynomial time
- [ ] Are essential to define the complexity class $\mathbf{NP}$

Your answer is partially correct.

Hai selezionato correttamente 1.
Le risposte corrette sono: If working in polynomial time, can be used to characterize $\mathbf{NP}$, Can be simulated by deterministic TMs.

Domanda **3**
Risposta corretta
Punteggio ottenuto 2,00 su 2,00
Contrassegna domanda

The universal Turing machine:

Scegli una o più alternative:

- [x] Can simulate every Turing machine, with a polynomial overhead. ✔
- [ ] Can simulate every Turing machine, but not itself
- [ ] Works in polynomial time.
- [x] Is an essential ingredient of in the proof of existence of uncomputable problems. ✔

Your answer is correct.

Le risposte corrette sono: Can simulate every Turing machine, with a polynomial overhead., Is an essential ingredient of in the proof of existence of uncomputable problems.

Domanda **4**
Risposta errata
Punteggio ottenuto 0,00 su 2,00
Contrassegna domanda

Suppose a language $\mathcal{L}$ is in $\mathbf{EXP}$ but not in $\mathbf{P}$. Then:

Scegli una o più alternative:

- [ ] $\mathcal{L}$ is necessarily $\mathbf{NP}$-complete.
- [x] The classes $\mathbf{NP}$ and $\mathbf{P}$ are different. ✖
- [x] There could be a nondeterministic polytime TM computing $\mathcal{L}$ ✔
- [ ] $\mathcal{L}$ can be computed in polynomial time.

Your answer is incorrect.

Le risposte corrette sono: $\mathcal{L}$ can be computed in polynomial time., There could be a nondeterministic polytime TM computing $\mathcal{L}$

Domanda **5**
Risposta corretta
Punteggio ottenuto 2,00 su 2,00
Contrassegna domanda

The notion of PAC-learnable concept class:

Scegli una o più alternative:

- [ ] Requires the output concept to have probability of error ε, in all cases
- [x] Does not make any reference to the time complexity of the learning algorithm ✔
- [x] Needs to hold for every distribution $\mathbf{D}$ on the instance class. ✔
- [ ] Cannot be reached when the underlying concept class is the one conjunctions of literals.

Your answer is correct.

Le risposte corrette sono: Needs to hold for every distribution $\mathbf{D}$ on the instance class., Does not make any reference to the time complexity of the learning algorithm